



National  
Qualifications  
2014

**X757/75/11**

**Physics  
Relationships Sheet**

THURSDAY, 22 MAY

9:00 AM – 11:00 AM



$$E_p = mgh$$

$$d = vt$$

$$E_k = \frac{1}{2}mv^2$$

$$v = f\lambda$$

$$Q = It$$

$$T = \frac{1}{f}$$

$$V = IR$$

$$A = \frac{N}{t}$$

$$R_T = R_1 + R_2 + \dots$$

$$D = \frac{E}{m}$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$H = Dw_R$$

$$V_2 = \left( \frac{R_2}{R_1 + R_2} \right) V_s$$

$$\dot{H} = \frac{H}{t}$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$s = vt$$

$$P = \frac{E}{t}$$

$$d = \bar{v}t$$

$$P = IV$$

$$s = \bar{v}t$$

$$P = I^2 R$$

$$a = \frac{v-u}{t}$$

$$P = \frac{V^2}{R}$$

$$W = mg$$

$$E_h = cm\Delta T$$

$$F = ma$$

$$p = \frac{F}{A}$$

$$E_w = Fd$$

$$E_h = ml$$

$$\frac{pV}{T} = \text{constant}$$

$$p_1V_1 = p_2V_2$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

# Additional Relationships

## Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

## Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

## Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

## Electron Arrangements of Elements

Group 1      Group 2

(1)

|          |          |     |
|----------|----------|-----|
| 1        | <b>H</b> |     |
| 1        |          |     |
| Hydrogen |          | (2) |

|         |           |           |           |
|---------|-----------|-----------|-----------|
| 3       | <b>Li</b> | 4         | <b>Be</b> |
| 2, 1    |           | 2, 2      |           |
| Lithium |           | Beryllium |           |

|         |           |           |           |
|---------|-----------|-----------|-----------|
| 11      | <b>Na</b> | 12        | <b>Mg</b> |
| 2, 8, 1 |           | 2, 8, 2   |           |
| Sodium  |           | Magnesium |           |

|            |          |            |           |
|------------|----------|------------|-----------|
| 19         | <b>K</b> | 20         | <b>Ca</b> |
| 2, 8, 8, 1 |          | 2, 8, 8, 2 |           |
| Potassium  |          | Calcium    |           |

|                |           |                |           |
|----------------|-----------|----------------|-----------|
| 37             | <b>Rb</b> | 38             | <b>Sr</b> |
| 2, 8, 18, 8, 1 |           | 2, 8, 18, 8, 2 |           |
| Rubidium       |           | Strontium      |           |

|                    |           |                    |           |
|--------------------|-----------|--------------------|-----------|
| 55                 | <b>Cs</b> | 56                 | <b>Ba</b> |
| 2, 8, 18, 18, 8, 1 |           | 2, 8, 18, 18, 8, 2 |           |
| Caesium            |           | Barium             |           |

|                        |           |                        |           |
|------------------------|-----------|------------------------|-----------|
| 87                     | <b>Fr</b> | 88                     | <b>Ra</b> |
| 2, 8, 18, 32, 18, 8, 1 |           | 2, 8, 18, 32, 18, 8, 2 |           |
| Francium               |           | Radium                 |           |

### Key

|                      |
|----------------------|
| Atomic number        |
| Symbol               |
| Electron arrangement |
| Name                 |

### Transition Elements

|                        |                         |                         |                         |                         |                         |                         |                         |                         |                         |
|------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| (3)                    | (4)                     | (5)                     | (6)                     | (7)                     | (8)                     | (9)                     | (10)                    | (11)                    | (12)                    |
| 21                     | 22                      | 23                      | 24                      | 25                      | 26                      | 27                      | 28                      | 29                      | 30                      |
| <b>Sc</b>              | <b>Ti</b>               | <b>V</b>                | <b>Cr</b>               | <b>Mn</b>               | <b>Fe</b>               | <b>Co</b>               | <b>Ni</b>               | <b>Cu</b>               | <b>Zn</b>               |
| 2, 8, 9, 2             | 2, 8, 10, 2             | 2, 8, 11, 2             | 2, 8, 13, 1             | 2, 8, 13, 2             | 2, 8, 14, 2             | 2, 8, 15, 2             | 2, 8, 16, 2             | 2, 8, 18, 1             | 2, 8, 18, 2             |
| Scandium               | Titanium                | Vanadium                | Chromium                | Manganese               | Iron                    | Cobalt                  | Nickel                  | Copper                  | Zinc                    |
| 39                     | 40                      | 41                      | 42                      | 43                      | 44                      | 45                      | 46                      | 47                      | 48                      |
| <b>Y</b>               | <b>Zr</b>               | <b>Nb</b>               | <b>Mo</b>               | <b>Tc</b>               | <b>Ru</b>               | <b>Rh</b>               | <b>Pd</b>               | <b>Ag</b>               | <b>Cd</b>               |
| 2, 8, 18, 9, 2         | 2, 8, 18, 10, 2         | 2, 8, 18, 12, 1         | 2, 8, 18, 13, 1         | 2, 8, 18, 13, 2         | 2, 8, 18, 15, 1         | 2, 8, 18, 16, 1         | 2, 8, 18, 18, 0         | 2, 8, 18, 18, 1         | 2, 8, 18, 18, 2         |
| Yttrium                | Zirconium               | Niobium                 | Molybdenum              | Technetium              | Ruthenium               | Rhodium                 | Palladium               | Silver                  | Cadmium                 |
| 57                     | 72                      | 73                      | 74                      | 75                      | 76                      | 77                      | 78                      | 79                      | 80                      |
| <b>La</b>              | <b>Hf</b>               | <b>Ta</b>               | <b>W</b>                | <b>Re</b>               | <b>Os</b>               | <b>Ir</b>               | <b>Pt</b>               | <b>Au</b>               | <b>Hg</b>               |
| 2, 8, 18, 18, 9, 2     | 2, 8, 18, 32, 10, 2     | 2, 8, 18, 32, 11, 2     | 2, 8, 18, 32, 12, 2     | 2, 8, 18, 32, 13, 2     | 2, 8, 18, 32, 14, 2     | 2, 8, 18, 32, 15, 2     | 2, 8, 18, 32, 17, 1     | 2, 8, 18, 32, 18, 1     | 2, 8, 18, 32, 18, 2     |
| Lanthanum              | Hafnium                 | Tantalum                | Tungsten                | Rhenium                 | Osmium                  | Iridium                 | Platinum                | Gold                    | Mercury                 |
| 89                     | 104                     | 105                     | 106                     | 107                     | 108                     | 109                     | 110                     | 111                     | 112                     |
| <b>Ac</b>              | <b>Rf</b>               | <b>Db</b>               | <b>Sg</b>               | <b>Bh</b>               | <b>Hs</b>               | <b>Mt</b>               | <b>Ds</b>               | <b>Rg</b>               | <b>Cn</b>               |
| 2, 8, 18, 32, 18, 9, 2 | 2, 8, 18, 32, 32, 10, 2 | 2, 8, 18, 32, 32, 11, 2 | 2, 8, 18, 32, 32, 12, 2 | 2, 8, 18, 32, 32, 13, 2 | 2, 8, 18, 32, 32, 14, 2 | 2, 8, 18, 32, 32, 15, 2 | 2, 8, 18, 32, 32, 17, 1 | 2, 8, 18, 32, 32, 18, 1 | 2, 8, 18, 32, 32, 18, 2 |
| Actinium               | Rutherfordium           | Dubnium                 | Seaborgium              | Bohrium                 | Hassium                 | Meitnerium              | Darmstadtium            | Roentgenium             | Copernicium             |

|                     |                     |                     |                     |                     |                     |
|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| (13)                | (14)                | (15)                | (16)                | (17)                | (18)                |
| 5                   | 6                   | 7                   | 8                   | 9                   | 10                  |
| <b>B</b>            | <b>C</b>            | <b>N</b>            | <b>O</b>            | <b>F</b>            | <b>Ne</b>           |
| 2, 3                | 2, 4                | 2, 5                | 2, 6                | 2, 7                | 2, 8                |
| Boron               | Carbon              | Nitrogen            | Oxygen              | Fluorine            | Neon                |
| 13                  | 14                  | 15                  | 16                  | 17                  | 18                  |
| <b>Al</b>           | <b>Si</b>           | <b>P</b>            | <b>S</b>            | <b>Cl</b>           | <b>Ar</b>           |
| 2, 8, 3             | 2, 8, 4             | 2, 8, 5             | 2, 8, 6             | 2, 8, 7             | 2, 8, 8             |
| Aluminium           | Silicon             | Phosphorus          | Sulfur              | Chlorine            | Argon               |
| 31                  | 32                  | 33                  | 34                  | 35                  | 36                  |
| <b>Ga</b>           | <b>Ge</b>           | <b>As</b>           | <b>Se</b>           | <b>Br</b>           | <b>Kr</b>           |
| 2, 8, 18, 3         | 2, 8, 18, 4         | 2, 8, 18, 5         | 2, 8, 18, 6         | 2, 8, 18, 7         | 2, 8, 18, 8         |
| Gallium             | Germanium           | Arsenic             | Selenium            | Bromine             | Krypton             |
| 49                  | 50                  | 51                  | 52                  | 53                  | 54                  |
| <b>In</b>           | <b>Sn</b>           | <b>Sb</b>           | <b>Te</b>           | <b>I</b>            | <b>Xe</b>           |
| 2, 8, 18, 18, 3     | 2, 8, 18, 18, 4     | 2, 8, 18, 18, 5     | 2, 8, 18, 18, 6     | 2, 8, 18, 18, 7     | 2, 8, 18, 18, 8     |
| Indium              | Tin                 | Antimony            | Tellurium           | Iodine              | Xenon               |
| 81                  | 82                  | 83                  | 84                  | 85                  | 86                  |
| <b>Tl</b>           | <b>Pb</b>           | <b>Bi</b>           | <b>Po</b>           | <b>At</b>           | <b>Rn</b>           |
| 2, 8, 18, 32, 18, 3 | 2, 8, 18, 32, 18, 4 | 2, 8, 18, 32, 18, 5 | 2, 8, 18, 32, 18, 6 | 2, 8, 18, 32, 18, 7 | 2, 8, 18, 32, 18, 8 |
| Thallium            | Lead                | Bismuth             | Polonium            | Astatine            | Radon               |

### Lanthanides

|                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| 57                 | 58                 | 59                 | 60                 | 61                 | 62                 | 63                 | 64                 | 65                 | 66                 | 67                 | 68                 | 69                 | 70                 | 71                 |
| <b>La</b>          | <b>Ce</b>          | <b>Pr</b>          | <b>Nd</b>          | <b>Pm</b>          | <b>Sm</b>          | <b>Eu</b>          | <b>Gd</b>          | <b>Tb</b>          | <b>Dy</b>          | <b>Ho</b>          | <b>Er</b>          | <b>Tm</b>          | <b>Yb</b>          | <b>Lu</b>          |
| 2, 8, 18, 18, 9, 2 | 2, 8, 18, 20, 8, 2 | 2, 8, 18, 21, 8, 2 | 2, 8, 18, 22, 8, 2 | 2, 8, 18, 23, 8, 2 | 2, 8, 18, 24, 8, 2 | 2, 8, 18, 25, 8, 2 | 2, 8, 18, 25, 9, 2 | 2, 8, 18, 27, 8, 2 | 2, 8, 18, 28, 8, 2 | 2, 8, 18, 29, 8, 2 | 2, 8, 18, 30, 8, 2 | 2, 8, 18, 31, 8, 2 | 2, 8, 18, 32, 8, 2 | 2, 8, 18, 32, 9, 2 |
| Lanthanum          | Cerium             | Praseodymium       | Neodymium          | Promethium         | Samarium           | Europium           | Gadolinium         | Terbium            | Dysprosium         | Holmium            | Erbium             | Thulium            | Ytterbium          | Lutetium           |

### Actinides

|                        |                         |                        |                        |                        |                        |                        |                        |                        |                        |                        |                        |                        |                        |                        |
|------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| 89                     | 90                      | 91                     | 92                     | 93                     | 94                     | 95                     | 96                     | 97                     | 98                     | 99                     | 100                    | 101                    | 102                    | 103                    |
| <b>Ac</b>              | <b>Th</b>               | <b>Pa</b>              | <b>U</b>               | <b>Np</b>              | <b>Pu</b>              | <b>Am</b>              | <b>Cm</b>              | <b>Bk</b>              | <b>Cf</b>              | <b>Es</b>              | <b>Fm</b>              | <b>Md</b>              | <b>No</b>              | <b>Lr</b>              |
| 2, 8, 18, 32, 18, 9, 2 | 2, 8, 18, 32, 18, 10, 2 | 2, 8, 18, 32, 20, 9, 2 | 2, 8, 18, 32, 21, 9, 2 | 2, 8, 18, 32, 22, 9, 2 | 2, 8, 18, 32, 24, 8, 2 | 2, 8, 18, 32, 25, 8, 2 | 2, 8, 18, 32, 25, 9, 2 | 2, 8, 18, 32, 27, 8, 2 | 2, 8, 18, 32, 28, 8, 2 | 2, 8, 18, 32, 29, 8, 2 | 2, 8, 18, 32, 30, 8, 2 | 2, 8, 18, 32, 31, 8, 2 | 2, 8, 18, 32, 32, 8, 2 | 2, 8, 18, 32, 32, 9, 2 |
| Actinium               | Thorium                 | Protactinium           | Uranium                | Neptunium              | Plutonium              | Americium              | Curium                 | Berkelium              | Californium            | Einsteinium            | Fermium                | Mendelevium            | Nobelium               | Lawrencium             |